

```
1 package designpatternsandprinciple.ques11;  
2  
3 public interface CustomerRepository {  
4  
5     String findCustomerById(int id);  
6 }
```

```
1 package designpatternsandprinciple.ques11;
2 public class CustomerRepositoryImpl implements CustomerRepository {
3
4     @Override
5     public String findCustomerById(int id) {
6
7         return "Customer ID: " + id + ", Name: Ambika";
8     }
9 }
```

```
1 package designpatternsandprinciple.ques11;
2
3 public class CustomerService {
4
5     private CustomerRepository repository;
6
7     public CustomerService(CustomerRepository repository) {
8         this.repository = repository;
9     }
10
11     public String getCustomer(int id) {
12         return repository.findCustomerById(id);
13     }
14 }
```

```
1 package designpatternsandprinciple.ques11;
2
3 public class DependencyInjectionTest {
4
5     public static void main(String[] args) {
6
7         CustomerRepository repository =
8             new CustomerRepositoryImpl();
9
10        CustomerService service =
11            new CustomerService(repository);
12
13        System.out.println(service.getCustomer(101));
14    }
15 }
```

Customer ID: 101, Name: Ambika